1. INTRODUCTION

In today's highly competitive subscription-based market, customer retention plays a crucial role in sustaining long-term business growth. Boxify, an online subscription box service delivering monthly lifestyle products, has observed a noticeable increase in customer churn over the past year. A significant number of subscribers are cancelling their subscriptions before completing six months, leading to revenue loss and reduced customer lifetime value.

This project aims to analyze historical customer data to identify the key factors contributing to churn and to support data-driven decision-making. By leveraging exploratory data analysis and interactive dashboards built in Power BI, the study seeks to uncover churn patterns, identify high-risk customer segments, and recommend actionable retention strategies aligned with business objectives.

The analysis strictly adheres to the constraints provided: using only the last three years of internal data, assuming monthly subscription cycles, and measuring churn only at the end of subscription periods.

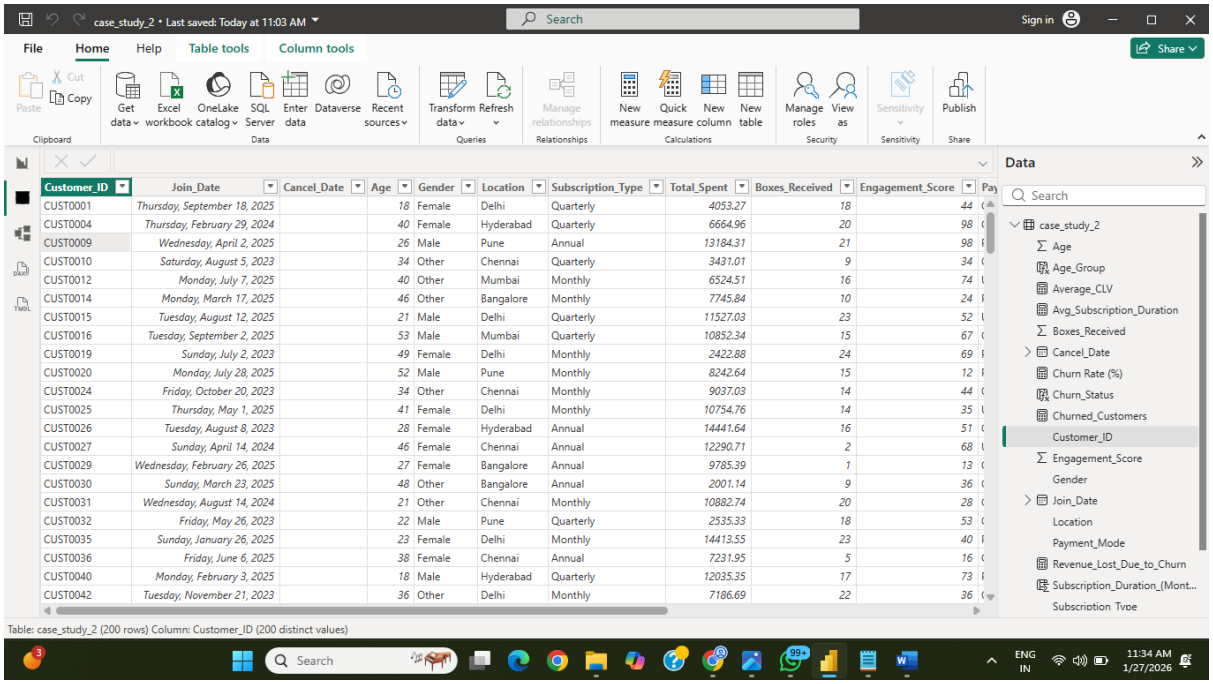
2. EXPLORATORY DATA ANALYSIS (EDA)

2.1 Dataset Overview

The dataset contains customer-level subscription information for the past three years, including:

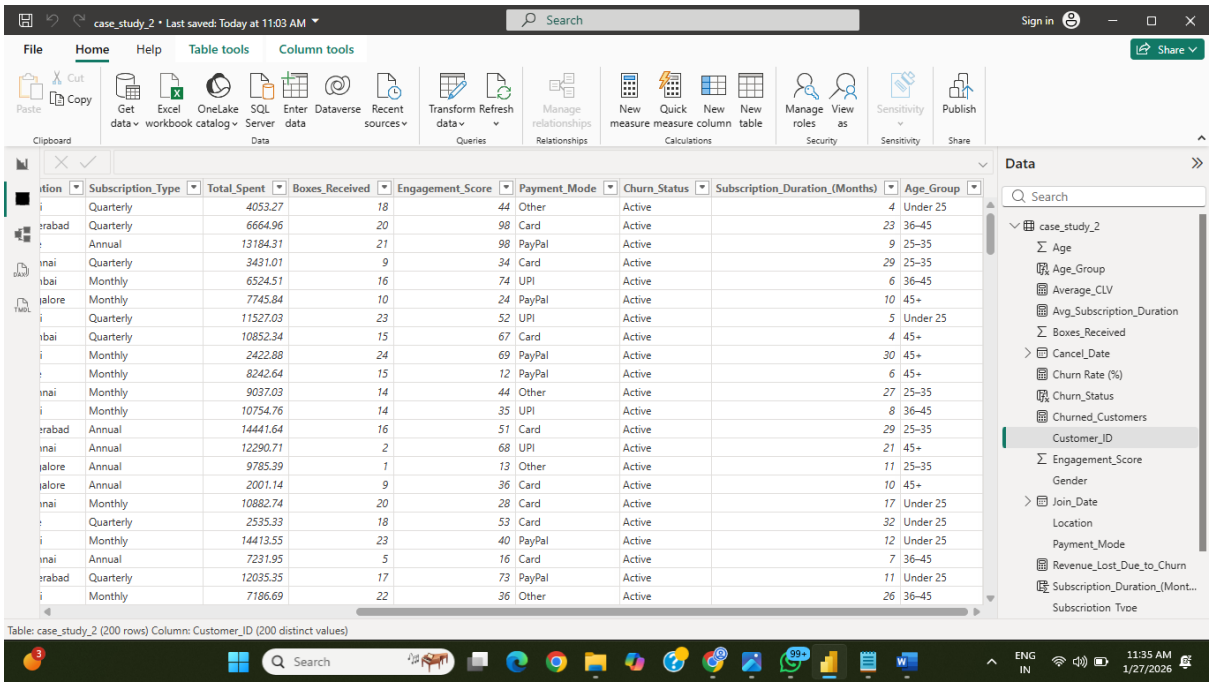
- Customer ID
- Age group
- Location
- Subscription type
- Subscription start and cancel dates
- Monthly engagement metrics
- Revenue generated

After initial inspection, duplicate records were removed to ensure data integrity. Missing values were carefully evaluated and handled based on business logic. For example, blank Cancel Date values indicate active customers rather than missing data.



Customer_ID	Join_Date	Cancel_Date	Age	Gender	Location	Subscription_Type	Total_Spent	Boxes_Received	Engagement_Score
CUST0001	Thursday, September 18, 2025		18	Female	Delhi	Quarterly	4053.27	18	44
CUST0004	Thursday, February 29, 2024		40	Female	Hyderabad	Quarterly	6664.96	20	98
CUST0009	Wednesday, April 2, 2025		26	Male	Pune	Annual	13184.31	21	98
CUST0010	Saturday, August 5, 2023		34	Other	Chennai	Quarterly	3431.01	9	34
CUST0012	Monday, July 7, 2025		40	Other	Mumbai	Monthly	6524.51	16	74
CUST0014	Monday, March 17, 2025		46	Other	Bangalore	Monthly	7745.84	10	24
CUST0015	Tuesday, August 12, 2025		21	Male	Delhi	Quarterly	11527.03	23	52
CUST0016	Tuesday, September 2, 2025		53	Male	Mumbai	Quarterly	10852.34	15	67
CUST0019	Sunday, July 2, 2023		49	Female	Delhi	Monthly	2422.88	24	69
CUST0020	Monday, July 28, 2025		52	Male	Pune	Monthly	8242.64	15	12
CUST0024	Friday, October 20, 2023		34	Other	Chennai	Monthly	9037.03	14	44
CUST0025	Thursday, May 1, 2025		41	Female	Delhi	Monthly	10754.76	14	35
CUST0026	Tuesday, August 8, 2023		28	Female	Hyderabad	Annual	14441.64	16	51
CUST0027	Sunday, April 14, 2024		46	Female	Chennai	Annual	12290.71	2	68
CUST0029	Wednesday, February 26, 2025		27	Female	Bangalore	Annual	9785.39	1	13
CUST0030	Sunday, March 23, 2025		48	Other	Bangalore	Annual	2001.14	9	36
CUST0031	Wednesday, August 14, 2024		21	Other	Chennai	Monthly	10882.74	20	28
CUST0032	Friday, May 26, 2023		22	Male	Pune	Quarterly	2535.33	18	53
CUST0035	Sunday, January 26, 2025		23	Female	Delhi	Monthly	14413.55	23	40
CUST0036	Friday, June 6, 2025		38	Female	Chennai	Annual	7231.95	5	16
CUST0040	Monday, February 3, 2025		18	Male	Hyderabad	Quarterly	12035.35	17	73
CUST0042	Tuesday, November 21, 2023		36	Other	Delhi	Monthly	7186.69	22	36

Table: case_study_2 (200 rows) Column: Customer_ID (200 distinct values)



Subscription_Type	Total_Spent	Boxes_Received	Engagement_Score	Payment_Mode	Churn_Status	Subscription_Duration_(Months)	Age_Group
Quarterly	4053.27	18	44	Other	Active	4	Under 25
Quarterly	6664.96	20	98	Card	Active	23	36–45
Annual	13184.31	21	98	PayPal	Active	9	25–35
Quarterly	3431.01	9	34	Card	Active	29	25–35
Monthly	6524.51	16	74	UPI	Active	6	36–45
Monthly	7745.84	10	24	PayPal	Active	10	45+
Quarterly	11527.03	23	52	UPI	Active	5	Under 25
Quarterly	10852.34	15	67	Card	Active	4	45+
Monthly	2422.88	24	69	PayPal	Active	30	45+
Monthly	8242.64	15	12	PayPal	Active	6	45+
Monthly	9037.03	14	44	Other	Active	27	25–35
Monthly	10754.76	14	35	UPI	Active	8	36–45
Annual	14441.64	16	51	Card	Active	29	25–35
Annual	12290.71	2	68	UPI	Active	21	45+
Annual	9785.39	1	13	Other	Active	11	25–35
Annual	2001.14	9	36	Card	Active	10	45+
Monthly	10882.74	20	28	Card	Active	17	Under 25
Quarterly	2535.33	18	53	Card	Active	32	Under 25
Monthly	14413.55	23	40	PayPal	Active	12	Under 25
Annual	7231.95	5	16	Card	Active	7	36–45
Quarterly	12035.35	17	73	PayPal	Active	11	Under 25
Monthly	7186.69	22	36	Other	Active	26	36–45

Table: case_study_2 (200 rows) Column: Customer_ID (200 distinct values)

2.2 Summary Statistics

Key summary metrics derived during EDA include:

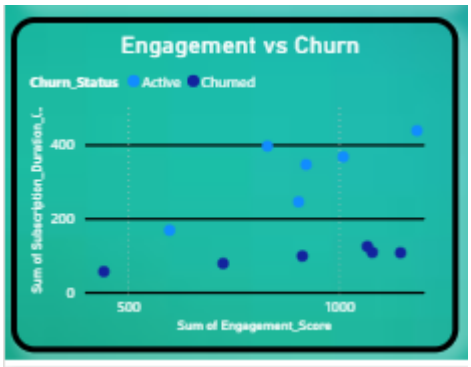
- Total number of customers

- Overall churn rate

- Average subscription duration (months)

- Average customer lifetime value (CLV)

- Average engagement rate per month

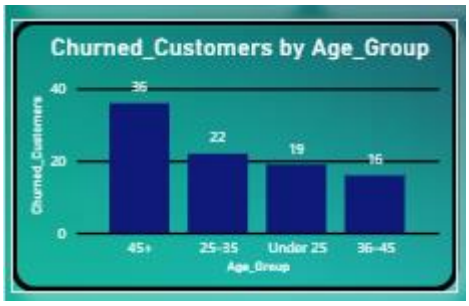


These metrics provided a high-level understanding of customer behavior and established a baseline for deeper analysis.

2.3 Churn Distribution Analysis

Churn was analyzed across multiple dimensions:

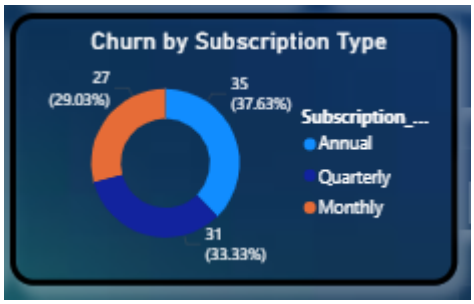
- Age group: Younger customer segments showed relatively higher churn.



- Location: Certain regions exhibited consistently higher cancellation rates.

- Subscription type: Monthly and basic plans experienced more churn compared to premium plans.

Bar charts and column visuals were used to clearly compare churn rates across segments.



3. METHODS

3.1 Churn Definition

A customer is classified as:

- **Churned**: If a cancel date exists
- **Active**: If the cancel date is blank

This logic aligns with the assumption that churn can only be measured at the end of a subscription cycle.

3.2 KPI Calculations

Key KPIs were calculated using DAX measures in Power BI:

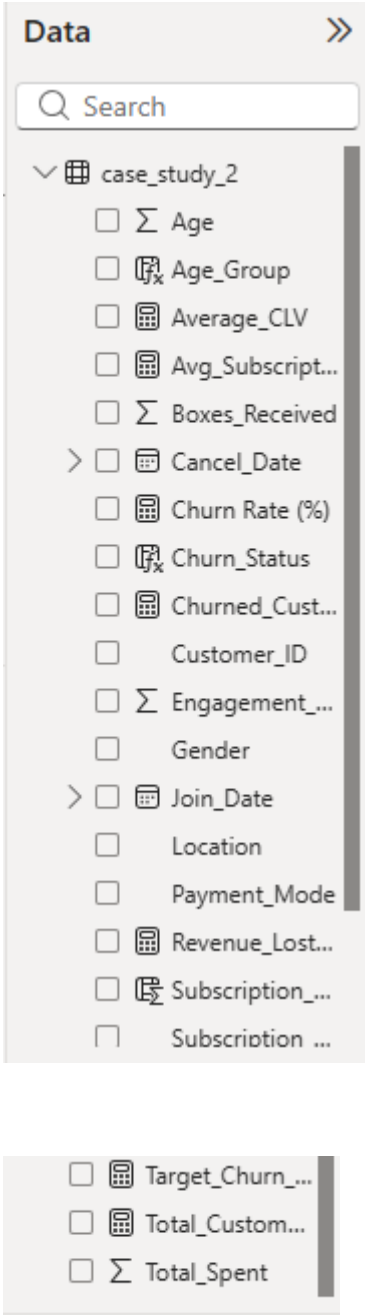
- **Churn Rate (%)**
  Percentage of customers who cancelled their subscription.
- **Customer Lifetime Value (CLV)**
  Total revenue generated by a customer before cancellation.
- **Engagement Rate**
  Average monthly interactions or logins.
- **Target Churn Rate (%)**
  Derived as a 10% reduction from the current churn rate, aligned with management's business goal.

3.3 Tools Used

- **Power BI**: Data modeling, DAX calculations, dashboard creation
- **Excel/CSV**: Initial data inspection and cleaning

4. DASHBOARD & RESULTS

An interactive Power BI dashboard was developed to visualize KPIs and answer business questions effectively.


Data
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Boxes_Received
Cancel_Date
Churn Rate (%)
Churn_Status
Churned_Cust...
Customer_ID
Engagement_...
Gender
Join_Date
Location
Payment_Mode
Revenue_Lost...
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Subscription ...
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Total_Custom...
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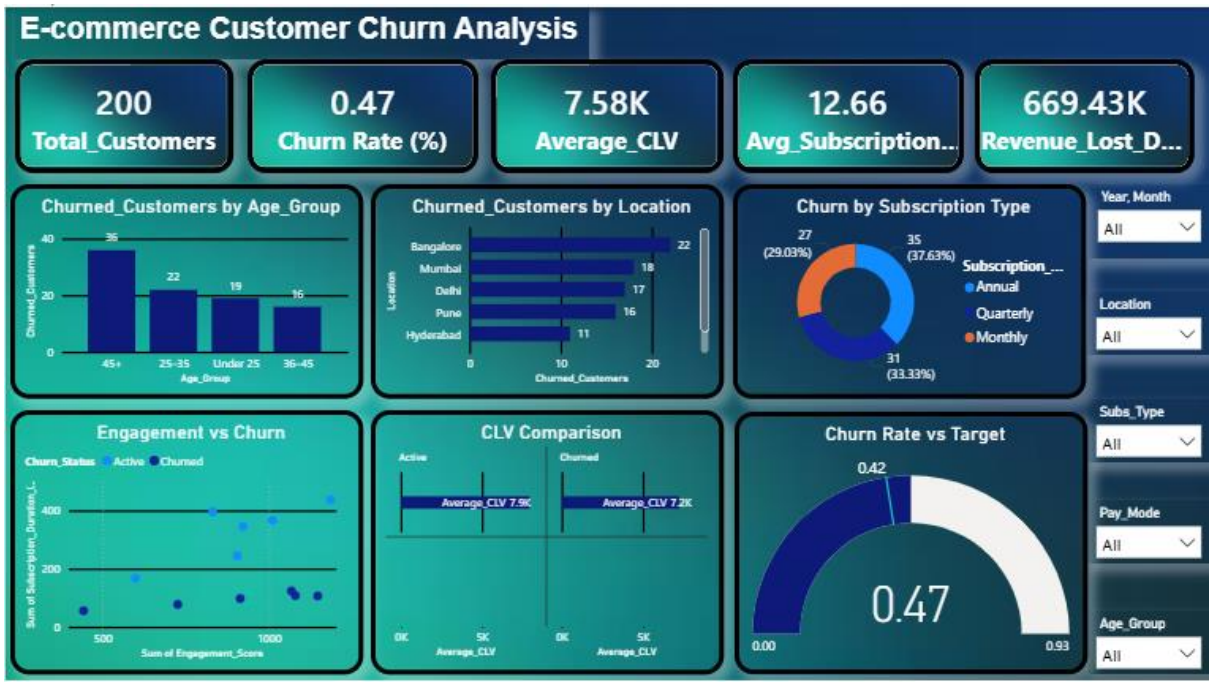


4.1 KPI Cards

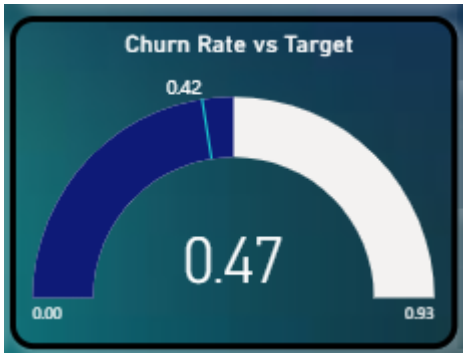
The dashboard includes KPI cards displaying:

- Total Customers
- Churn Rate %
- Average CLV
- Average Subscription Duration
- Revenue Lost Due to Churn

These KPIs dynamically update based on slicer selections.

4.2 Gauge Chart

A gauge chart titled **"Churn Rate vs Target"** compares the current churn rate against the target churn rate (10% reduction). This visualization allows management to instantly assess whether retention goals are being met.



4.3 Segmentation Visuals

The dashboard includes:

- Churn by Age Group
- Churn by Location
- Churn by Subscription Type

Dropdown slicers enable filtering by time period, customer segment, and subscription plan, ensuring flexibility and interactivity.

5. INSIGHTS

Key insights derived from the analysis include:

- Customers with **low engagement levels** are significantly more likely to churn.
- Shorter subscription durations (less than 6 months) show the highest churn rates.
- Basic subscription plans experience more churn compared to premium plans.
- Certain locations consistently show higher churn, suggesting regional or logistical issues.
- Churned customers have a noticeably **lower average CLV** compared to active customers.

These insights highlight the strong relationship between engagement, subscription value, and retention.

6. RECOMMENDATIONS

6.1 Quick Wins (Short-Term)

- Introduce personalized discounts for customers with declining engagement.
- Trigger retention emails or offers during the first 3–5 months of subscription.
- Improve onboarding experience to increase early engagement.

6.2 Long-Term Strategies

- Develop loyalty programs for long-term subscribers.
- Enhance premium subscription benefits to encourage upgrades.
- Use engagement-based segmentation for targeted marketing campaigns.
- Continuously monitor churn KPIs through the dashboard.

6.3 High-Risk Customer Engagement

- Identify customers with low engagement and short subscription duration.
- Offer customized boxes or content recommendations.
- Implement feedback surveys before cancellation.

7. CONCLUSIONS

This analysis successfully identified key churn drivers at Boxify using historical subscription data. By combining exploratory data analysis with interactive dashboards, the study provided actionable insights into customer behavior and churn patterns.

The findings indicate that improving customer engagement, enhancing early subscription experience, and targeting high-risk segments can significantly reduce churn. If implemented effectively, the recommended strategies can help Boxify achieve its goal of reducing churn by 5–10% over the next six months.

The dashboard serves as a powerful decision-support tool, enabling stakeholders to track performance, evaluate strategies, and make data-driven decisions.

8. REFERENCES

1. Boxify Internal Customer Dataset (Last 3 Years)
2. Microsoft Power BI Documentation

3. Subscription Business Retention Best Practices
4. Customer Lifetime Value Analysis Concepts